

Vision Based Target Tracking Using Rotary Wing Vehicles

Mr Christopher Jones and Dr Thomas Richardson

University of Bristol, Department of Aerospace Engineering, Queen's Building, University Walk, Bristol, BS8 1TR

Abstract

The work presented in this paper is intended as a summary of the research being completed at The University of Bristol into the field of vision based target tracking using rotary wing vehicles. Work has been ongoing at the university in this field for two years with results and demonstrations presented at the 1st and 2nd SEAS DTC conferences. This paper details the progress of the work over the past year toward achieving vision-based target tracking onboard a six degree of freedom rotary wing vehicle. A study has been completed showing the effect of target dimensions and how this influences the accuracy of the adopted vision system. Results are also presented showing the relative benefits of using visually derived data for target based station keeping over more traditional global positioning system techniques. Using vision as a real-time sensor incurs time latencies in image acquisition, processing and data transfer and results are presented showing the experimentally determined parameters that contribute to the overall system latency. Work has continued using the Quanser 3 degree of freedom rotary wing platform demonstrated at the 2nd SEAS DTC Conference in July 2007 and results are presented for both theoretical and experimental control system designs and implementations as well as adopted filtering techniques. A 6 degree of freedom rotary wing vehicle has been acquired through funding from the SEAS DTC for use with this research and initial flight testing using this vehicle has been undertaken. Results are presented for station-keeping using this craft and a conventional sensor feedback configuration. Target tracking functionality will be incorporated into the navigation strategy and details are presented regarding how this is to be achieved for demonstration at the 3rd SEAS DTC Conference in June 2008.

Keywords: Vision, Control, UAV, SR20, Quanser, 3 DoF, Target Tracking, TALEOS

Introduction

The work presented in this paper is intended as a summary of the research being carried out at The University of Bristol in the field of autonomous target tracking. The work is to be undertaken over 3 years, having begun in October 2005 and due for completion in September 2008. Funding for this research comes from the Systems Engineering for Autonomous Systems Defence Technology Centre (SEAS DTC), the University of Bristol Unmanned Aerial Vehicles (UAV) Conference and the Engineering and Physical Sciences Research Council (EPSRC).

The advancement in computing capabilities over the past decade has enhanced the possibility of using a range of new sensory devices. One such application is the use of real-time vision processing to provide optically interpreted information that can be used by control system architectures. The use of vision systems onboard unmanned vehicles ranges from using insect inspired optical flow velocity interpretation for flying in urban canyons, to processing complex scenes and extracting features in order to aid vehicle guidance. The main benefit of using vision as a sensory device is accredited to the ability to correct for inertial drift in Global

Positioning System (GPS) denied environments [1] although sensor time penalties are incurred due to image capture and processing latencies. It is envisaged that a possible extension to the research being conducted could include this augmentation.

One of the primary design factors when using a vision system as a sensory device is the option to use monocular or binocular/stereo vision. Monocular vision systems simplify the image interpretation algorithms, and reduce computational demands, but are unable to provide pose estimations in a single time step without *a priori* knowledge of the target configuration. This is a resolvable criteria when using binocular vision, given knowledge of the distance between the capture devices, although there will be an inherent delay introduced as a result of the additional computation required. The choice between using monocular and binocular approaches is at the discretion of the control system designer, with examples available for both systems using a range of flight vehicles [2, 3, 4].

Using a vision system onboard an unmanned vehicle can be approached from the perspective of inner-loop craft stabilisation or outer-loop guidance augmentation. The work outlined in this paper has made use of a monocular vision sensor for both reduced order craft stabilisation techniques as well as outer loop guidance proposals onboard on a free-flight rotary wing vehicle.

The final aim of the research is the experimental demonstration of translating target tracking using a free-flight rotary wing vehicle and work is ongoing to achieve this goal. The work presented in this paper details the accuracy of the vision solution being used in addition to quantifying the various aspects of the sensor latency. Comparisons are made between conventional GPS based solutions

and vision alternatives, with clear boundaries ascertained for the use of each approach.

Work has continued using a 3 Degree of Freedom (DoF) helicopter to perform inner-loop craft stabilisation and guidance based upon visually interpreted sensor data. An outline is proposed for using this equipment to quantify the operational envelope, and craft capabilities, using vision-based control architectures. The envelope boundaries will be defined by the vision system capabilities, the control system design and the craft dynamics.

As part of the ongoing research, the SEAS DTC has funded the acquisition of a free-flight 6 DoF rotary wing vehicle. This craft was displayed at the 2nd SEAS DTC Conference in July 2007 and since that time has been experimentally tested. Flight trials to date include the operation of the craft over a standard 35MHz radio frequency link under manual control and over a 2.4GHz wireless link via the Automatic Flight Control System (AFCS) to provide station-keeping utilising a conventional sensor array. An outline is presented detailing how this system will be augmented to include the use of vision-based guidance strategies to enable autonomous translating target tracking suitable for demonstration at the 3rd SEAS DTC Conference in June 2008.

Vision System Evaluation

As part of the ongoing research into using vision in the control loop for rotary wing vehicles, it is necessary to quantify the performance and accuracy of the sensor. To achieve this, a bespoke testing rig was designed and constructed as seen in Figure 1.

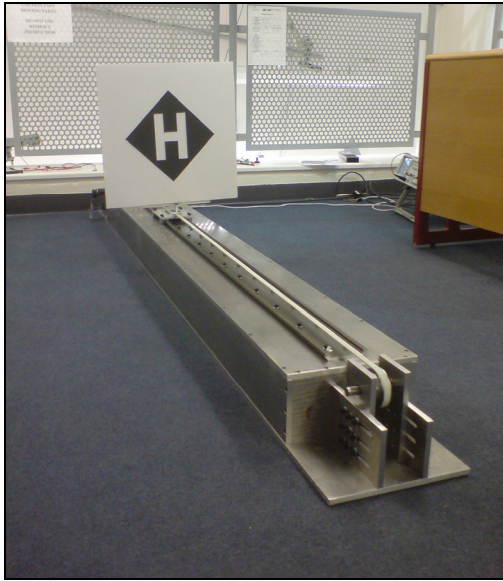


Figure 1: Vision Testing Equipment

The vision software to be evaluated is entitled Terrain Aided Localisation using Electro-Optical Sensors (TALEOS) and is provided by BAE Systems, Filton. The software uses a Canny edge detection technique in conjunction with model matching to estimate 6 DoF camera pose relative to the observed model [5]. This pose estimation can then be interpreted and used accordingly, in this instance directly in the control loop for vision based navigation. For the purposes of the system evaluation, only translational position estimations are considered with the axes as defined in Figure 2.

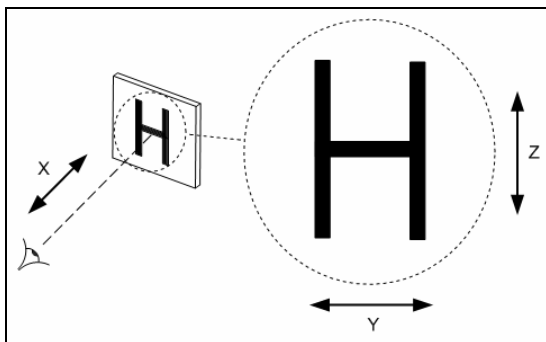


Figure 2: TALEOS Axes

To fully exploit the capabilities of the vision sensor, it is necessary to understand and quantify the accuracy constraints and so experimentation using the linear rig seen in Figure 1 was undertaken. The

performance characteristics of TALEOS are predominantly defined by the scale of the target within the image plane and the distance it is from the imaging device. To ascertain a relationship between the size of the target and the minimum and maximum distances as which a lock on the target can be achieved, four target sizes were considered and linear relationships established, as seen in Figure 3.

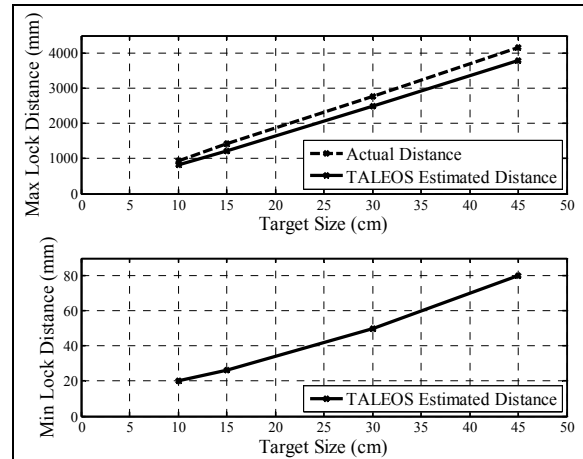


Figure 3: Min/Max Lock Distances

The minimum and maximum lock distances are both linear functions of the target size and thus it is possible to extrapolate and evaluate a required target size for a given minimum or maximum distance consideration. There is however a discrepancy in the TALEOS measurement of distance, the actual distance estimated by the vision software is a constant 12% less than the absolute, physically measured position. This can however be corrected through the use of a fixed scaling factor.

A significant parameter of any sensory device is the accuracy under given conditions. To assess the performance of the vision sensor, an estimation of the accuracy at varying distances is required. Testing was performed to assess the standard deviation of the TALEOS X measurement at set distances over a given time frame for varying target sizes. The variation in position estimation was recorded over 45 seconds at each target

distance and the mean and variance of this data was calculated.

To assess the influence of target size, the area of the target, in square pixels, within the image plane was considered. It was expected that a small target at a small distance would appear with the same dimensions in the image plane as a large target at a large distance and thus the accuracy of both was anticipated to be similar. Figure 4 highlights the approximated variations in this target area with distance from the camera for differing target sizes.

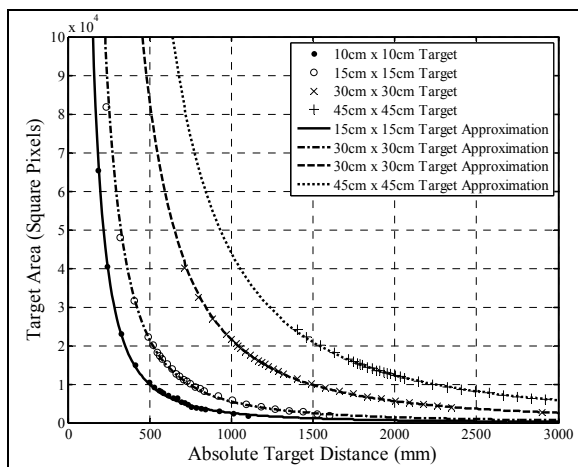


Figure 4: Target Area Approximation

By assessing the standard deviation, and normalising with respect to the distance between the camera and the target, it was possible to approximate the relationship between the error measure (y) and the target area (x) as the exponential function:

$$y = 1738 x^{-1.34}$$

This worst case approximation, and original experimental results, can be seen in Figure 5.

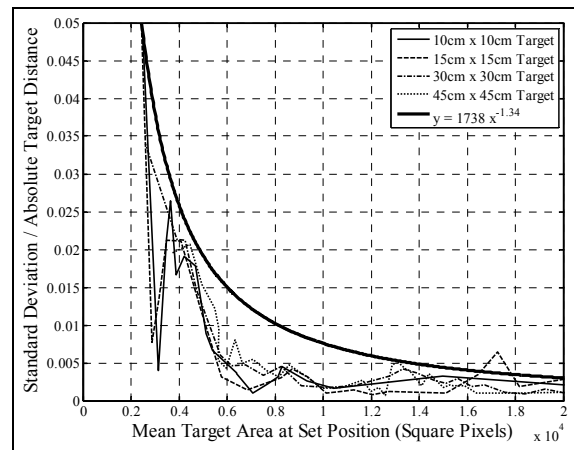


Figure 5: Standard Deviation Approximation

It is thus possible to estimate the envisaged error in sensor performance given a target area within the image plane. It is of more practical use to consider the relationship in terms of target size metrics. By making several graphical approximations and restructuring of the data, this can be achieved as seen in Figure 6.

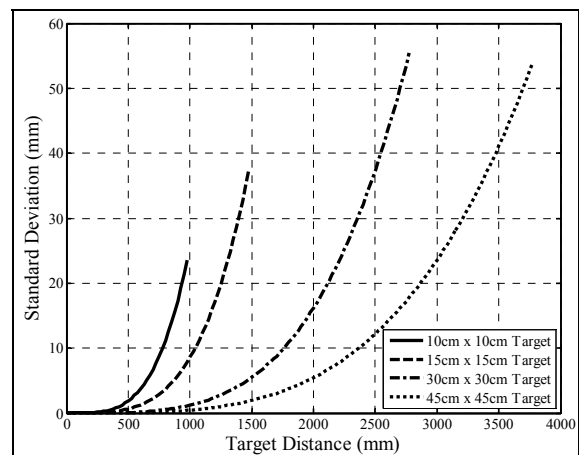


Figure 6: TALEOS Error Schematic

Figure 6 highlights the total anticipated error in measurement for a given target size at a given distance and enables the expected performance of the vision system to be evaluated. The remaining aspect of the study is to remove the necessity of prior knowledge of the target size and produce schematics for a generic consideration such that it is possible to directly establish the expected error for any sized target, not just the four considered here.

Most unmanned rotary wing vehicles use GPS at least in part for control, guidance or navigation. Problems with this approach are common in GPS denied environments, such as urban canyons, and vision systems offer a viable alternative to this conventional approach.

GPS and Differential Global Positioning System (DGPS) have an associated constant error parameter in each axis whereas vision solutions are enhanced at close range, as highlighted in Figure 6. For this reason, the method adopted for comparison is to examine the relative accuracies normalised with regard to the total distance to a target, as seen in Figure 7.

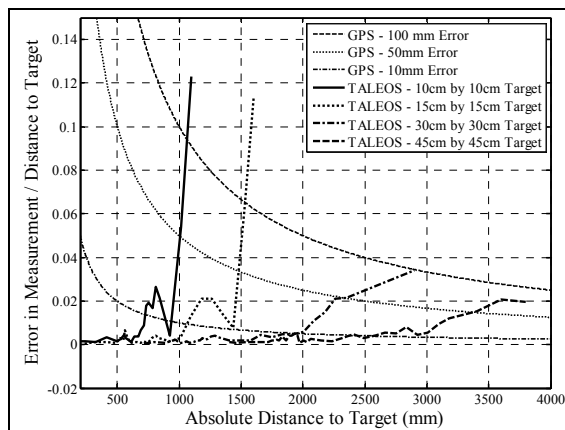


Figure 7: GPS vs. Vision Trade-off

It is possible to ascertain from Figure 7 the regions in which using a vision system for craft navigation is beneficial in comparison to a variety of GPS solutions. It is clear that when using a 15cm by 15cm target, vision is a more accurate tracking solution over a 50mm positional error in GPS up to distances of 1500mm. By plotting such results, it is possible for a control system designer to plan architecture switching based around this information and factor target size and distance into system design from an early stage.

Any sensory device will usually incur latency in measurement, whether the sensor is rapid or slow, and will operate with varying accuracies. It is essential to be able

to identify any system latencies and the use of a visual sensor incurs a quantifiable time delay which needs to be established.

The process of using vision in the control loop has several associated processes from image capture to the control system demand calculation, each stage of which has measurable time latency. These delays applied to a generic rotary wing platform can be summarised as:

- 1) Image Capture
- 2) Image Transfer
- 3) TALEOS Processing
- 4) TALEOS to Control System
- 5) Control Architecture Processing
- 6) Control System to Flight Vehicle

Each of these delays can be said to be fixed or variable, and this governs the overall time budget associated with this control structure. It is possible through a variety of measurement techniques to establish values for all these processes.

The image capture and transfer is a function of the imaging device in question and in this research, an AXIS 207W wireless network camera was used and operated over an 802.11g wireless network. By capturing and time stamping an image frame whilst the camera is facing the display screen, as seen in Figure 8, it was possible to determine the total image capture and transfer time, in this instance 130ms.

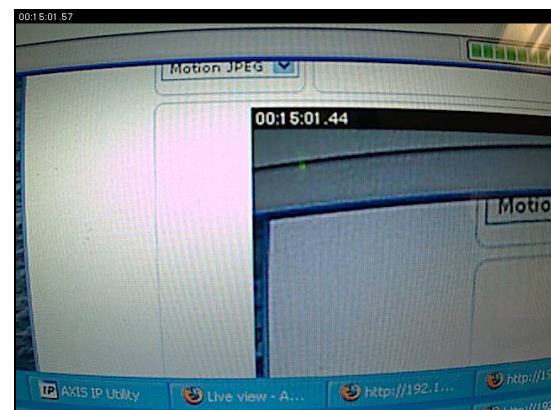


Figure 8: Image Capture Latency

The received image is then passed to the TALEOS vision software for pose estimation. Parameters were inserted into the operating loop of the software enabling the processing time to be quantified.

The processing speed was seen to vary depending upon the operating point. When the target had been acquired and was being tracked by the software, speeds of 40Hz were observed whereas with no lock, this speed dropped to 20Hz. The total time latency is thus between 25 and 50ms.

All experimentation to date has utilised an RS232 communication interface for transferring data between the TALEOS processing computer and the separate control system processor. This data contains a variety of vision system parameters and is 8 Bytes in size. The RS232 connection parameters were set to transfer data at 9600 Bits/sec inducing a total transfer time of $9600 / (8 \times 8) \approx 7$ ms. This data is then processed by the control architecture and the processing time will be defined by the nature of the flight vehicle being considered. In this instance, a C program is to be utilised to process positional demands and details of this technique can be seen in a later section.

Transfer of data to the AFCS is an aspect which is again specific to the vehicle being considered. For this research, this communication takes place over a wireless network and though network Packet InterNet Grouper (PING) data transfer requests of suitable size. It was possible to estimate the data transfer time as ≈ 6 ms. All of these time delays from image capture to craft actuation can be expressed diagrammatically as seen in Figure 9.

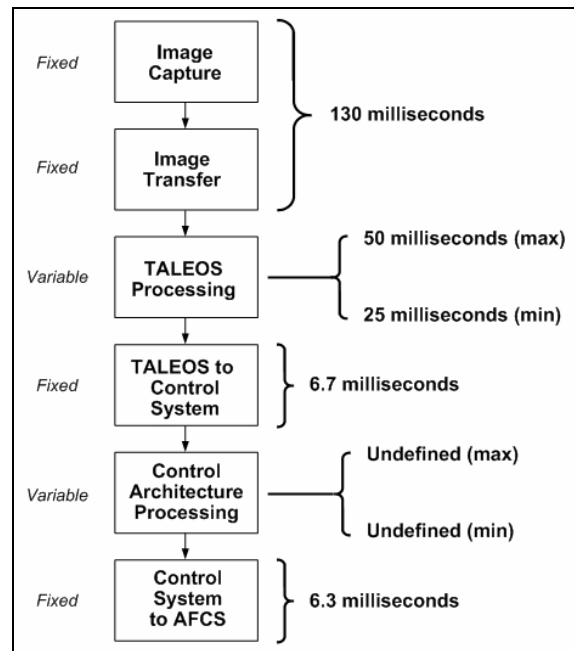


Figure 9: Time Latency Overview

The total time delay between image capture and control system actuation based on this analysis is between 168ms and 193ms (neglecting the control architecture processing time) with the majority of the latency arising from image capture and transfer.

The 6 DoF experimental flight vehicle being used in conjunction with this research is updated with positional demands over a wireless network connection at a rate of 4Hz. It is thus plausible that if the control architecture processing time is less than 57ms, the time delay will have little influence on the update rate to the AFCS.

Quanser 3 DoF Helicopter

The research being conducted has made use of a limited degree of freedom rotary wing vehicle supplied by Quanser®. The experimental equipment is a 3 DoF helicopter platform with a central slip-ring for unlimited rotation around the vertical axis. The rotation about the vertical axis is termed the travel axis, the height of the vehicle above the table, elevation and the rotation of the craft head, pitch. The equipment was presented at the 2nd SEAS

DTC Conference in July 2007 and can be seen in Figure 10.

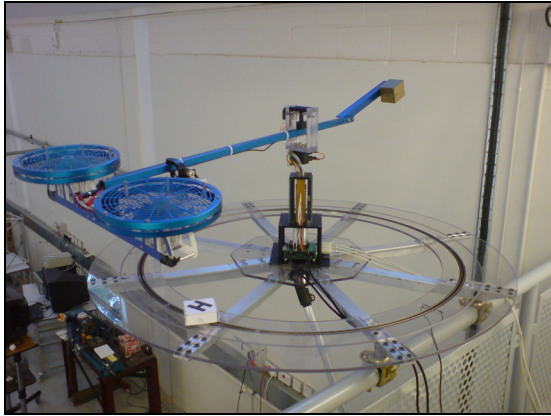


Figure 10: Quanser 3 DoF Helicopter

A demonstration was exhibited highlighting a vision-based stabilisation and target tracking problem using this equipment. Vision-based sensory data was utilised to maintain craft elevation and to provide positional demands in the travel axis to demonstrate vision-based control and guidance on a physical system.

The control architecture of this system took the form of Proportional, Integral, Differential (PID) control loops in the travel and elevation axes and Proportional Differential (PD) control in the pitch axis. Each of these gains was tuned using a Linear Quadratic Regulator (LQR) approach [6] and relevant gain terms extracted from the resultant matrix [7]. The LQR approach is an optimal control solution and requires a linear system representation in state-space format.

A theoretical model of the 3 DoF helicopter was obtained and verified through comparison with experimental data. The model used was a linearised state-space representation of the system, valid when the main arm of the equipment is in a horizontal position [8]. The original model was modified and restructured such that the states were represented as:

$$\begin{bmatrix} \text{Elevation} \\ \text{Pitch} \\ \text{Travel} \\ \text{Elevation Rate} \\ \text{Pitch Rate} \\ \text{Travel Rate} \\ \int \text{Elevation} \\ \int \text{Travel} \end{bmatrix}$$

Arbitrary initial Q and R weighting matrices were selected for controller tuning and LQR scheme 1 took the form:

$$Q = \begin{bmatrix} 1000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1000 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1000 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 100 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 100 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 100 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 10 \end{bmatrix}$$

$$R = \begin{bmatrix} 5 & 5 \end{bmatrix}$$

The weighting matrices were modified based upon the observed response characteristics until a suitable response was achieved using LQR scheme 2:

$$Q = \begin{bmatrix} 2500 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 600 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1200 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 600 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2000 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 100 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 900 \end{bmatrix}$$

$$R = \begin{bmatrix} 5 & 5 \end{bmatrix}$$

The improvement in response using these two control schemes can be seen in Figure 11.

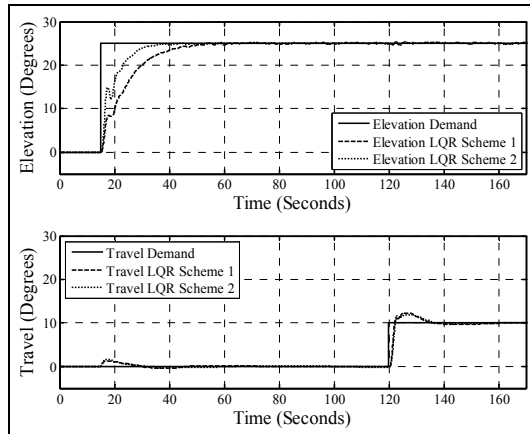


Figure 11: LQR Tuning Results

The result of this tuning process was to move the open-loop system eigenvalues to new locations and provide improved damping properties.

Open Loop	Closed Loop
$-0.1004 \pm 0.9513i$	$-0.1308 \pm 1.0063i$
$-0.1630 \pm 1.1813i$	$-0.2974 \pm 1.4021i$
-0.2517	$-0.0916 \pm 0.3290i$
0	-0.0232
0	-0.3451
0	

The validation of the theoretical model was achieved by analysing theoretical and experimental system state variations with identical input demands. This can be seen in Figure 12 for both the elevation and travel axes.

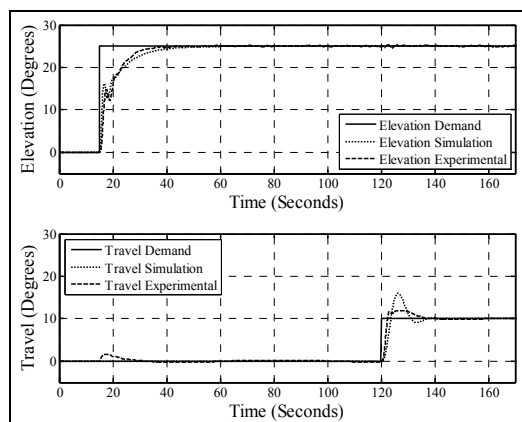


Figure 12: Simulation vs. Experimental Quanser 3 DoF Results

The simulated and experimental responses are seen to agree in both magnitude and damping in both of the axes considered. The pitch response is of a similar comparative nature, although results are not included here. Theoretical tuning of the control architecture was also completed using eigenstructure assignment making use of this experimentally verified linear system representation. This procedure was undertaken to verify the LQR method adopted and ensure the desired closed loop system properties were obtained. The results of this procedure demonstrated responses inline with those seen in Figure 12.

The vision system output is a set of parameters detailing the current pose of the camera relative to the target being tracked and it is these parameters that are utilised in the control architecture. An example image frame of the vision system in operation on-board the 3 DoF helicopter can be seen in Figure 13.

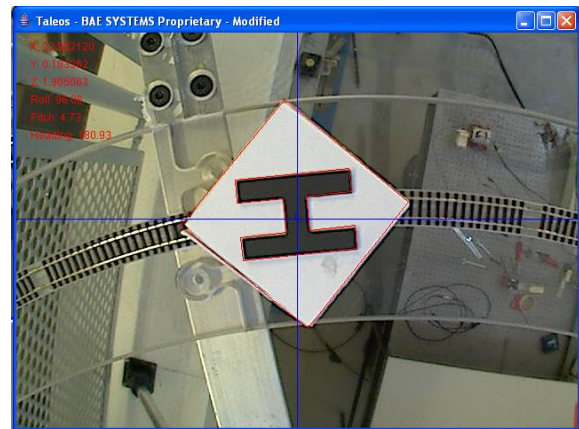


Figure 13: TALEOS in Operation

The accuracy and range details of this system have been highlighted in Figures 3 and 6 for a 10cm by 10cm target as used in this experimentation. It is possible to compare the TALEOS estimation of position to the reading from the equipment mounted optical encoders in both the elevation and travel axes (conversion from encoder measurements in degrees to elevation in cm is necessary for comparison). The result of this comparison

to a 5 degree step change in elevation and an 8 degree step change in travel can be seen in Figures 14 and 15 respectively.

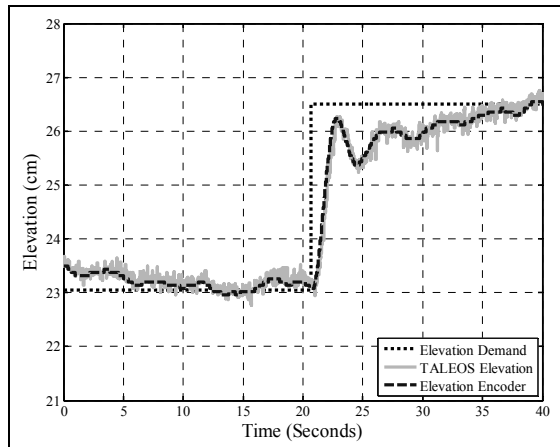


Figure 14: Vision Tracking Accuracy in Elevation

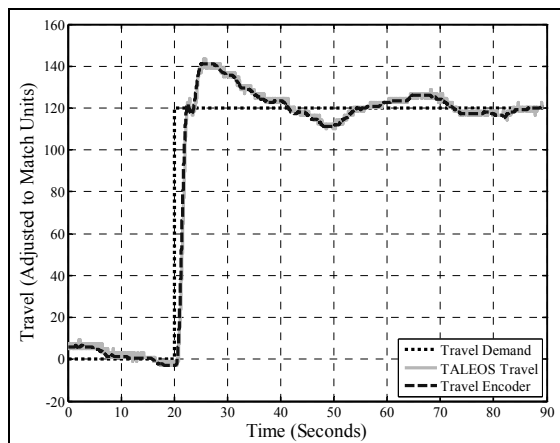


Figure 15: Vision Tracking Accuracy in Travel

Figures 14 and 15 highlight the vision system data accuracy in terms of mean value but suffer from estimation variations inline with the results seen in Figure 6. To improve this estimation variance, a Kalman filter was designed and implemented such that the measurement readings are analysed and assessed with relation to the theoretical dynamic model of the system with the resultant filtered data suffering from less variance.

By utilising the positional estimation of TALEOS in conjunction with the designed control system, it is possible to analyse the accuracy of the target tracking solution in maintaining position directly above the target being tracked. The accuracy in

tracking in the travel axis can be seen in Figure 16.

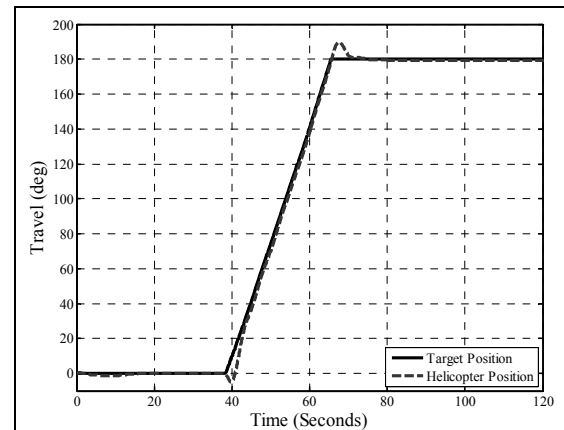


Figure 16: Helicopter Tracking Accuracy

There is an inherent inaccuracy in tracking at the 2 instances when the target velocity changes. In these cases, the craft is over/undershooting the target position due to the control system design. The vision-system data is however still being used for tracking and as such any overshoot is expressed in the TALEOS estimation of the position in the image frame. Comparison of this image frame positional estimation with the error in the helicopter tracking relative to the target position yields a response as seen in Figure 17.

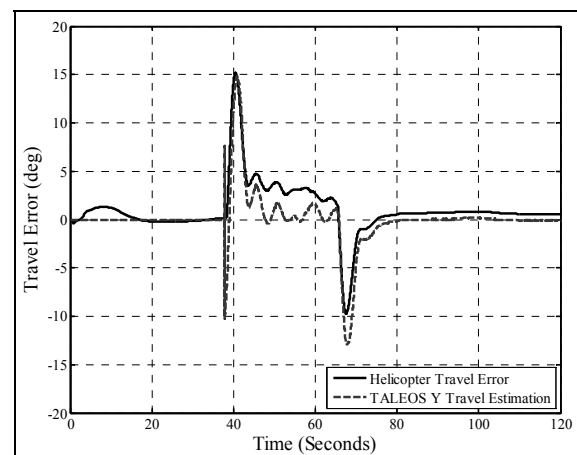


Figure 17: Helicopter Tracking Error vs. TALEOS Estimation

The TALEOS estimation of image frame position is observed to be inline with the current error in the craft travel and as expected, any variation away from zero error is corrected by the control system to

bring the craft directly above the target. The maximum error allowable in this scenario directly dictates the ability of the control solution to accurately track a target. Any deviation in excess of 20 degrees leads to the target leaving the image frame and visual lock is lost. This maximum error gives a direct measure of the control system capabilities and governs the potential applications of any solution.

The final Quanser control system architecture appears as seen in Figure 18, including the Kalman filter implementation.

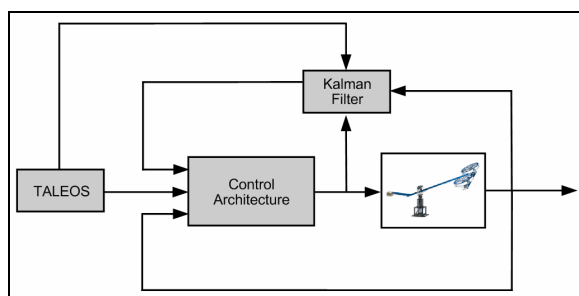


Figure 18: Quanser Control System

The 3 DoF equipment has been used to produce experimental results demonstrating the application of vision-based control to a reduced order flight vehicle. There remains the necessity to examine the system stability in the presence of increasing feedback latency such that the stability margins and robustness of the system can be examined. Further investigation will be completed toward using the dynamic vision model and artificially inserted time delays in the control system feedback loop to establish the operational envelope limitations based upon vision system constraints, vehicle dynamics and control system tuning parameters. A detailed set of operational parameters will be established with envelope boundaries represented quantitatively before verification is completed on a 6 DoF experimental platform.

6 DoF Rotary Wing Experimentation

The research being conducted has made use of a 6 DoF unmanned rotary wing flight vehicle, the SR20, supplied by Rotomotion^(LLC) and funded through the SEAS DTC. This craft is a commercial off-the-shelf solution and provides a stable platform for experimentation utilising vision-in-the-loop guidance and navigation. It is an electrically powered size 90 scale helicopter and has an endurance of 30 minutes, a 3kg payload capability and maximum forward flight velocity of 25 knots. The craft is also equipped with an on-board pan/tilt camera and is pictured in Figure 19.



Figure 19: SR20 6 DoF Test Platform

The vehicle is controlled via a radio frequency safety pilot model aircraft transmitter or alternatively through the built-in AFCS. All AFCS control demands are passed to the onboard processor through the supplied Ground Control Station (GCS), pictured in Figure 20.

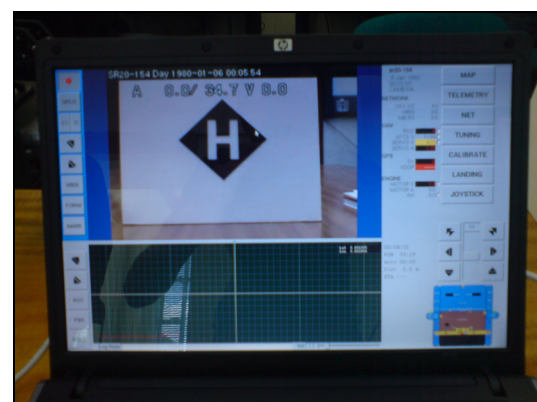


Figure 20: SR20 GCS – Laboratory Testing

Flight testing performed to date has included manual flight operation and autonomous station-keeping using the onboard AFCS and appropriate manually entered positional demands. The SR20 undergoing flight testing by The University of Bristol can be seen in Figure 21.



Figure 21: SR20 In-flight Testing

The AFCS is capable of receiving information either through the provided Graphical User Interface (GUI), via a Rotomotion Perl encoded program or through a custom written C code via Application Programming Interface (API) based commands. It is the last of these methods which is to be used to augment the guidance strategy to demonstrate target tracking capabilities. The intended approach is to utilise the C interface to receive target location data from TALEOS via an RS232 data link, read data from the SR20 onboard sensors via the API, and then compute positional demands based on this data. The resultant positional demands will then be passed to the GCS, and in-turn the AFCS, such that the process of tracking a translating target will be demonstrated. This process will be completed within the 57ms time limit identified in the time budget study. Experimentation will initially be carried out at altitudes in excess of 10 metres with a suitably sized target inline with the scalability results presented regarding target size and maximum lock distances. This work will be completed and presented at the 3rd SEAS DTC Conference in June 2008.

6 DoF Simulation

In parallel to the experimental work being completed onboard the 6 DoF free-flight vehicle, a theoretical representation of a similar sized vehicle has been investigated. A linear representation of the rotary wing vehicle was obtained from The University of Kansas [9]. Initial control system tuning results using this model were presented at the 2nd SEAS DTC Conference including simulated target tracking functionality [10]. It has been possible to build a virtual environment directly linked to the MATLAB simulation, via the Virtual Reality toolbox, to represent the craft responses in a 3 dimensional setting. An image of a target tracking scenario can be seen in Figure 22.

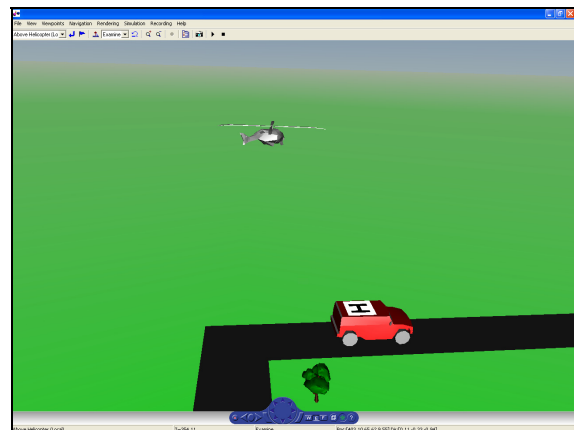


Figure 22: 6 DoF Virtual Environment

The simulation environment is to be utilised to investigate system stability in the presence of a dynamic vision model, established from the study presented in Figures 3 to 6. The limitations of vision-based control in 6 DoF in the presence of feedback latencies will be explored and presented at the SEAS DTC Conference 2008.

An area of interest highlighted within the SEAS DTC is that of gusting with regard to unmanned vehicles. For this reason, it has been proposed that a Dryden gust model is incorporated into the simulation with varying degrees of gust severity to examine the effect of turbulence with specific regard

to the target tracking problem being addressed in this research. Results of this study will be presented at the SEAS DTC Conference in June 2008.

Conclusions

This paper has served to highlight work that has been completed in the field of vision-based control and guidance and has outlined plans to complete a variety of demonstrations. A detailed analysis of the accuracy of the vision software has been presented in conjunction with a time latency study concerning delays using vision-in-the-loop control architectures. It has been possible to identify design criteria for using vision based localisation techniques over a more conventional GPS solution.

Theoretical and experimental results have been presented for both a reduced order 3 DoF and a free-flight 6 DoF rotary wing vehicle. Validation of the theoretical modelling has been completed through the experimental comparison of data with accurate results achieved.

The final aim of the research is to demonstrate autonomous target tracking using a 6 DoF rotary wing vehicle. This platform has now been acquired and flown and is ready for rapid development to enable this tracking functionality. A demonstration of this will be presented at the 3rd SEAS DTC Conference in June 2008.

Future Work

The research is focussed on several specific areas of interest and this paper has highlighted the work to be completed. Work stemming from a systems study into vision system accuracy will be incorporated into a simulation environment of a 6 DoF helicopter to enable the capabilities, and more specifically limitations, of using vision-based navigation to be assessed. This process of operational envelope

identification will be first examined through the simulation and experimental use of the 3 DoF rotary wing equipment and will involve artificial time latency insertion in the feedback loop to assess the system stability.

Owing to the interest of the SEAS DTC in gusting and the restrictions of flight vehicles in this environment, it has been proposed that a study be completed into the gust response of a rotary wing vehicle, and applied to this research, how this influences the target tracking functionality. Results of this study will be presented at the 3rd SEAS DTC Conference in June 2008.

The final aim of this research is almost complete with steps in place to achieve the goal of experimentally demonstrating translating target tracking from onboard a 6 DoF free-flight platform. A demonstration of this capability will be completed and presented at the 3rd SEAS DTC Conference.

References

- [1] S. Mehta, G. Hu, N. Gans, W. Dixon *Adaptive Vision-Based Collaborative Tracking Control of an UGV via a Moving Airborne Camera: A Daisy Chaining Approach*, Proceedings of the 45th IEEE Conference on Decision and Control, Pg 3867-3872, 2006.
- [2] F. Panerai, G. Metta, and G. Sandini, *Visuo-Inertial Stabilization in Space-Vibrant Binocular Systems*, Robotics and Autonomous Systems, Pg 195–214, 2000.
- [3] Z. He, R. Iyer, and P. Chandler, *Vision-Based UAV Flight Control and Obstacle Avoidance*, Proceedings of the 2006 American Control Conference, Pg 2166–2170, 2006.
- [4] V. Chitrakaran, D. Dawson, J. Chen, and H. Kannan, *Velocity and Structure Estimation of a Moving Object Using a Moving Monocular Camera*, Proceedings of the 2006 American Control Conference, Pg 5159–5164, 2006.
- [5] P. Collins, A. Stephens, P. Greenway, R. Deaves, M. Priestly and M. Bullen, *Terrain Aided Localisation Using Electro-Optical Sensing (TALEOS)*, SPIE Volume 3069, Pg 342–353, 1997.

- [6] B. Anderson and J. Moore, *Optimal Control: Linear Quadratic Methods*, Prentice Hall of India, 2004.
- [7] Quanser, 3 DoF Helicopter Experiment, www.quanser.com.
- [8] M. Ishutkina, *Design and Implementation of a Supervisory Safety Controller of a 3 DOF Helicopter*, Masters Thesis, Massachusetts Institute of Technology, June 2004.
- [9] S. Bhandari and R. Colgren *6-DoF Dynamic Model for a Raptor 50 UAV Helicopter Including Stabilizer Bar Dynamics*, AIAA Modelling and Simulation Technologies Conference and Exhibit, 2006.
- [10] C. Jones and T. Richardson, *Vision Based Tracking Using Rotary Wing Vehicles*, Proceedings of the 2nd SEAS DTC Conference, 2007.

Acknowledgements

The work reported in this paper was funded by the Systems Engineering for Autonomous Systems (SEAS) Defence Technology Centre established by the UK Ministry of Defence. Additional funding comes from the University of Bristol UAV Conference and the EPSRC. Acknowledgement is also given to BAE Systems, UK, for their ongoing and valuable assistance with the research.